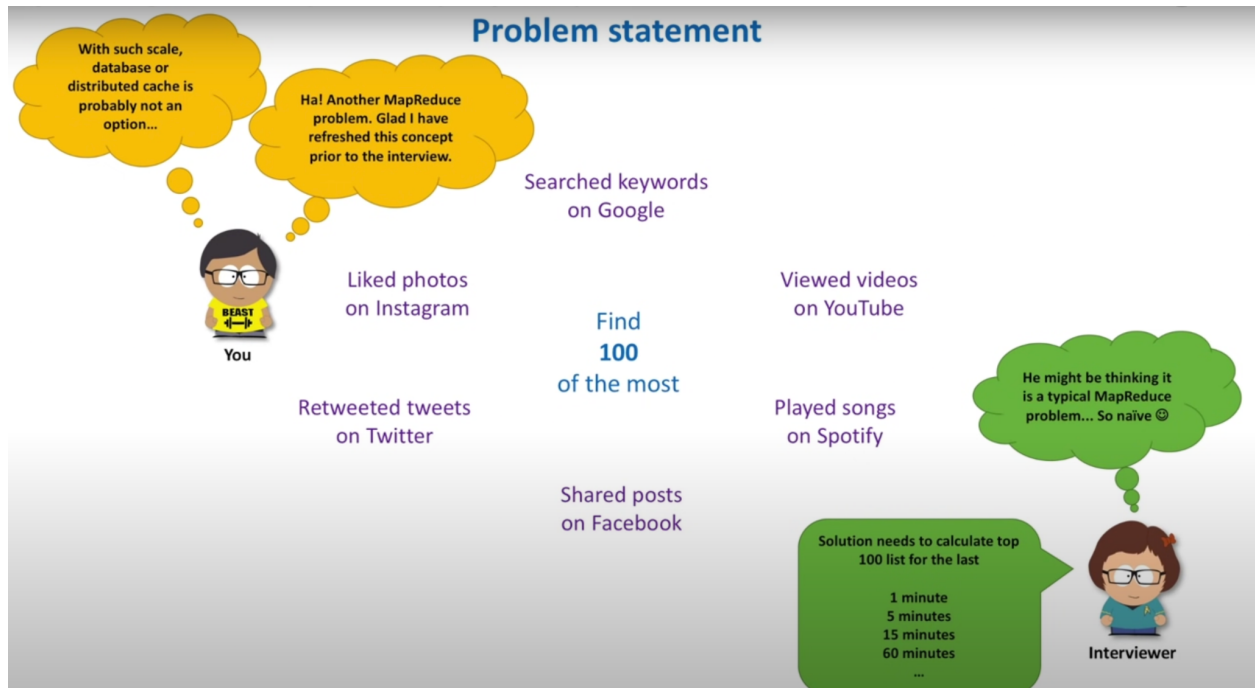


Top K Problem (Heavy Hitters)

▶ System Design Interview - Top K Problem (Heavy Hitters)



Requirements

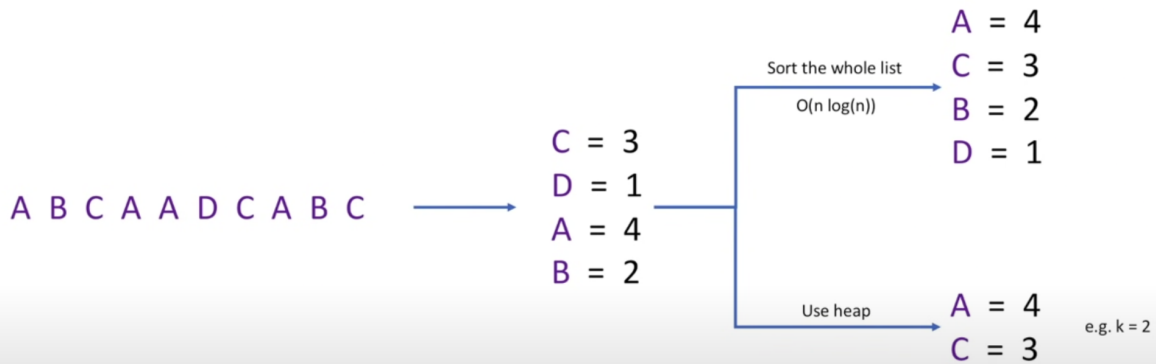
Functional

- $\text{topK}(k, \text{startTime}, \text{endTime})$

Non-Functional

- **Scalable** (scales out together with increasing amount of data: videos, tweets, posts, etc.)
- **Highly Available** (survives hardware/network failures, no single point of failure)
- **Highly Performant** (few tens of milliseconds to return top 100 list)
- **Accurate** (as accurate as we can get)

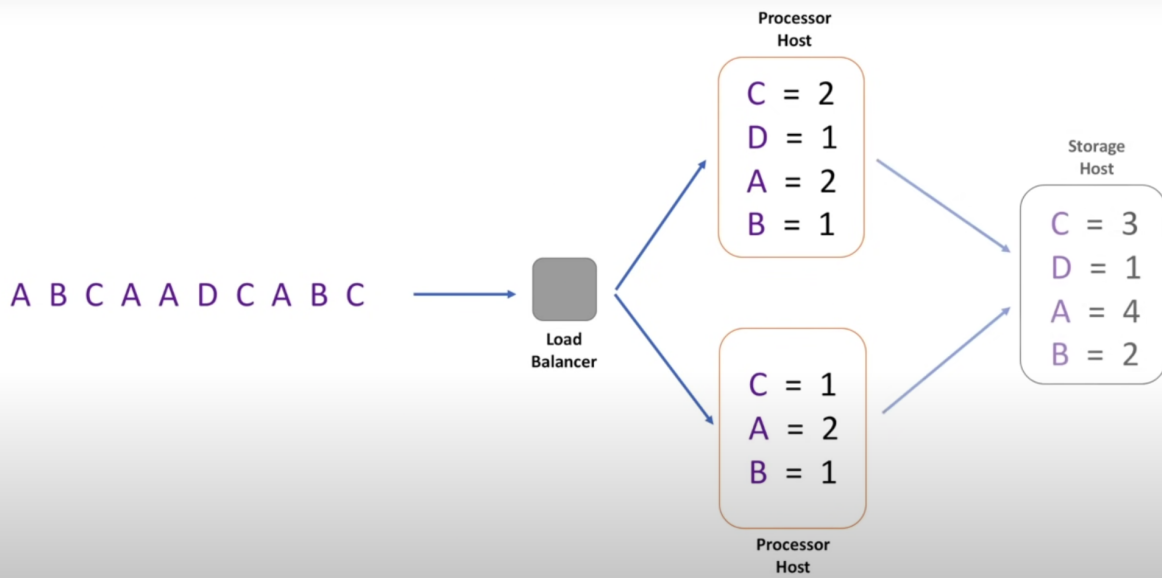
Hash table, single host



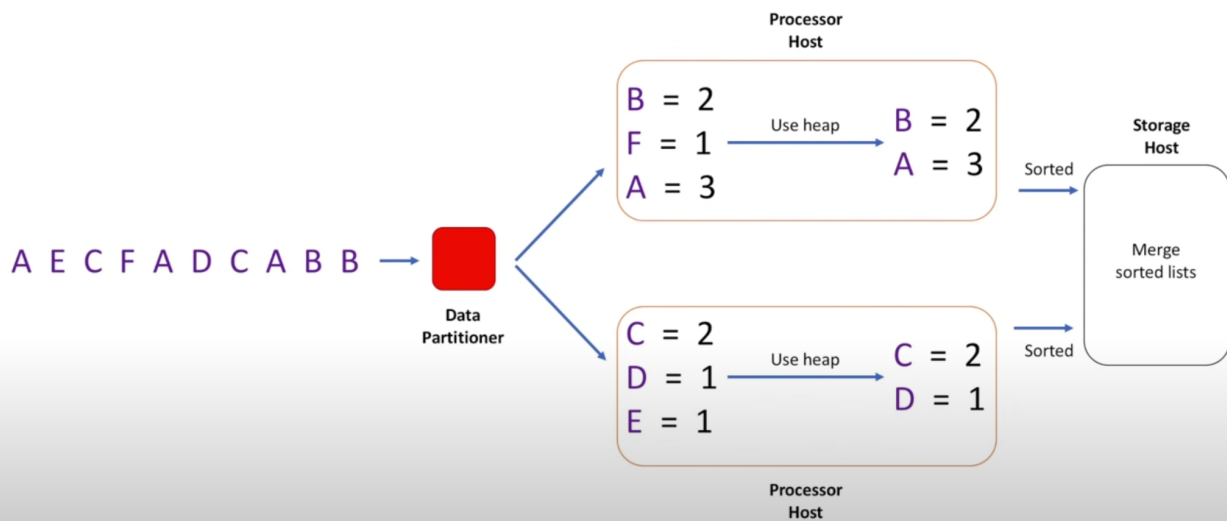
Top k algorithm implementation

```
public List<HeavyHitter> topK(String[] events, int k) {  
    Map<String, Integer> frequencyTable = new HashMap<String, Integer>();  
    for (String event : events) {  
        frequencyTable.put(event, frequencyTable.getOrDefault(event, 0) + 1);  
    }  
  
    PriorityQueue<HeavyHitter> heap =  
        new PriorityQueue<HeavyHitter>(Comparator.comparing(e -> e.getFrequency()));  
  
    for (Map.Entry<String, Integer> entry : frequencyTable.entrySet()) {  
        heap.offer(new HeavyHitter(entry.getKey(), entry.getValue()));  
  
        if (heap.size() > k) {  
            heap.poll();  
        }  
    }  
  
    List<HeavyHitter> result = new ArrayList<HeavyHitter>();  
    while (heap.size() > 0) {  
        result.add(heap.poll());  
    }  
  
    return result;  
}  
  
class HeavyHitter {  
    private final String identifier;  
    private final int frequency;  
  
    public HeavyHitter(String identifier, int frequency) {  
        this.identifier = identifier;  
        this.frequency = frequency;  
    }  
}
```

Hash table, multiple hosts



Hash table, multiple hosts, partitioning



Count-min sketch

A

<i>h5</i>				1		
<i>h4</i>					1	
<i>h3</i>	1					
<i>h2</i>			1			
<i>h1</i>		1				

width

Count-min sketch

A A

<i>h5</i>				2		
<i>h4</i>					2	
<i>h3</i>	2					
<i>h2</i>			2			
<i>h1</i>		2				

width

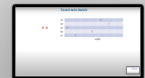


Count-min sketch

B A A A

<i>h5</i>				1	3		
<i>h4</i>						3	1
<i>h3</i>	3		1				
<i>h2</i>				3			1
<i>h1</i>		4					

↑
collision
width



Count-min sketch

C B A A A

<i>h5</i>	1			1	3		
<i>h4</i>			1			3	1
<i>h3</i>	3		1		1		
<i>h2</i>				3			2
<i>h1</i>		5					

↑
collision
width
←
collision



Count-min sketch

C B A A A

h5	1			1	3		
h4			1			3	1
h3	3		1		1		
h2				3			2
h1		5					

width

collision

collision

How is data retrieved?
For example, what is the returned value of A?

Minimum of all the counters for the element.
4

How do we choose width and height (number of hash functions)?

e - accuracy we want to have
 d - certainty with which we reach the accuracy
 $width = \lceil 2.0 / e \rceil$
 $height = \lceil -\log(1 - d) / \log(2) \rceil$

Count-min sketch returns frequency count estimate for a single item.
Where is a top-k list in this picture?

Keep a heap of top-k elements.

High-level architecture

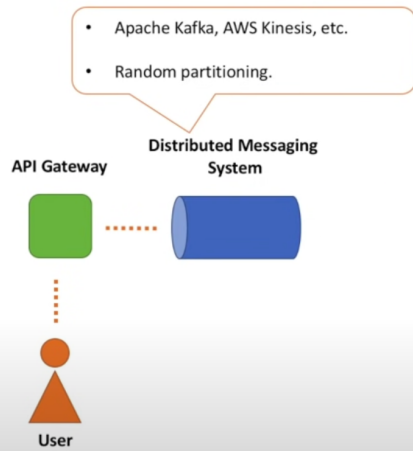
- Single entry point for all clients.
- Aggregates data on the fly or via a background process that processes logs. Data is flushed based on either time or size.
- Serializes data in a compact binary format (e.g. Apache Avro).

API Gateway

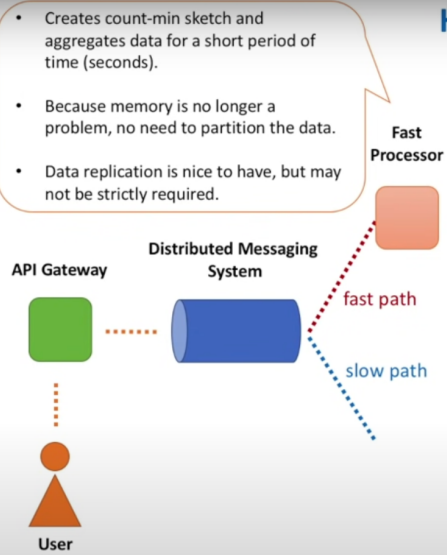


User

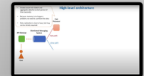
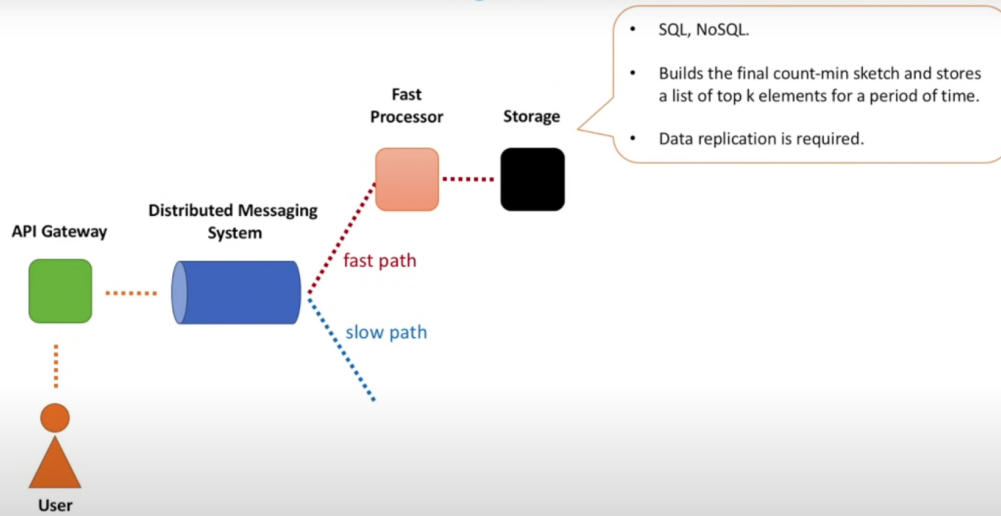
High-level architecture



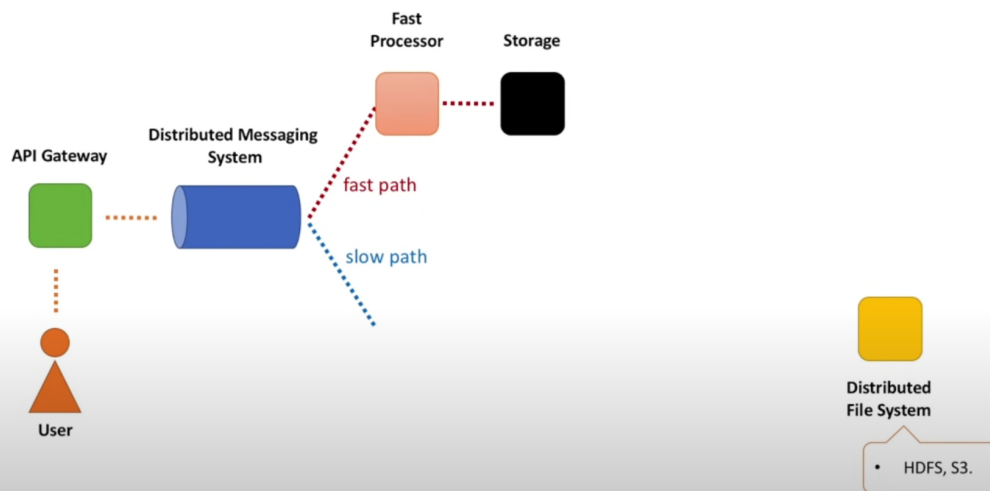
High-level architecture



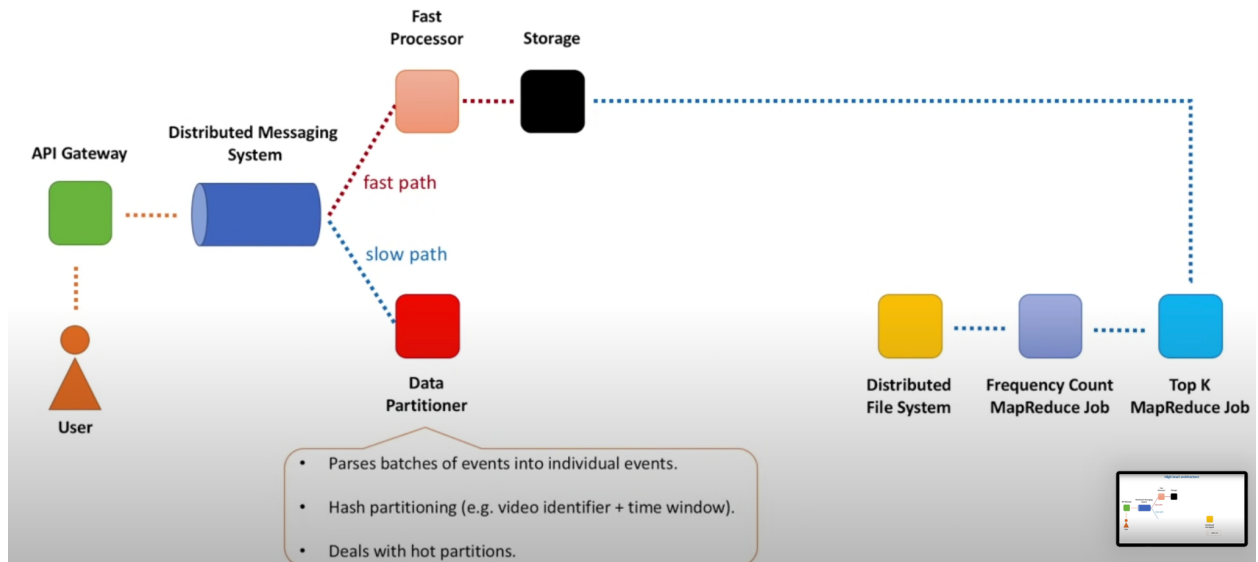
High-level architecture



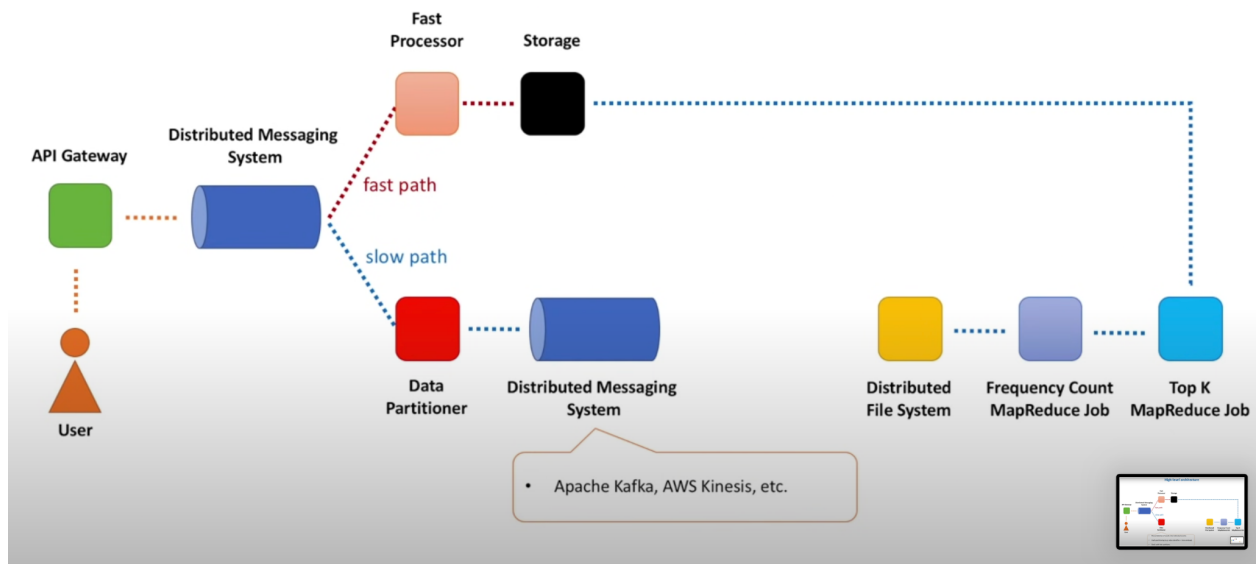
High-level architecture



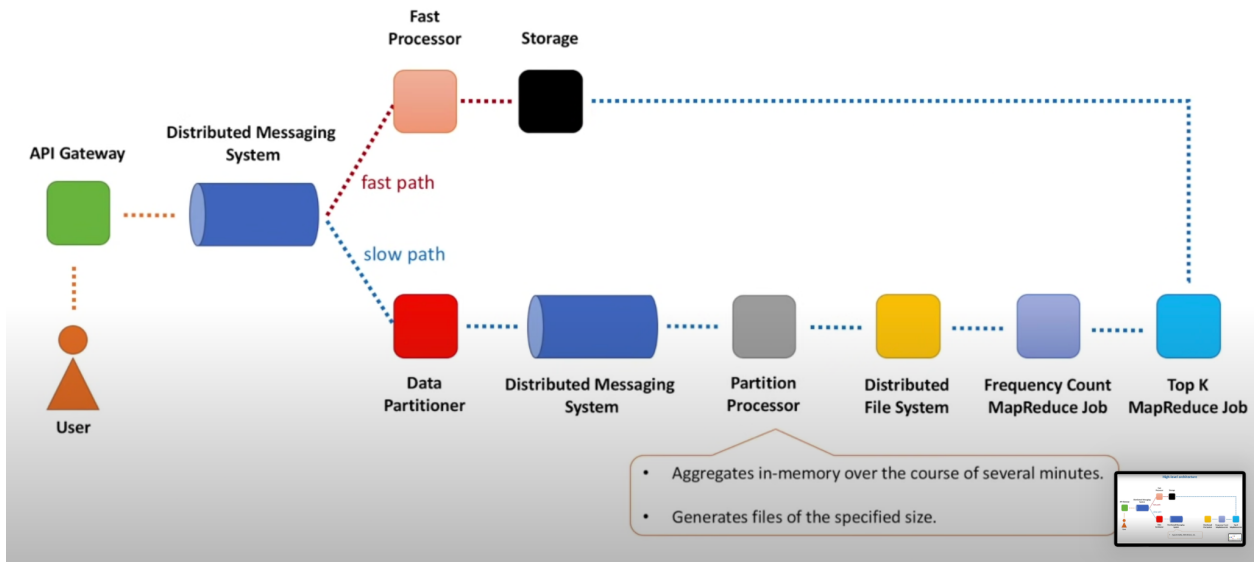
High-level architecture



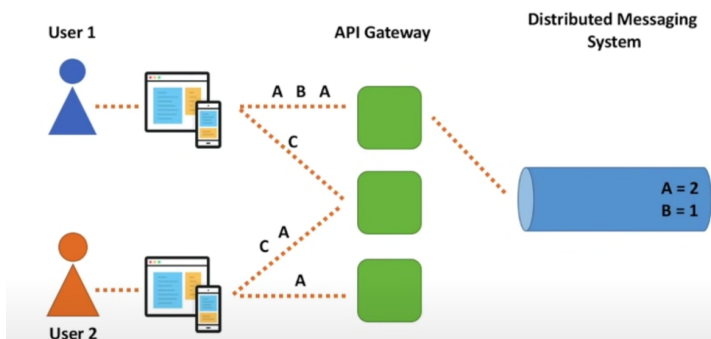
High-level architecture



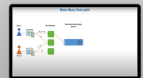
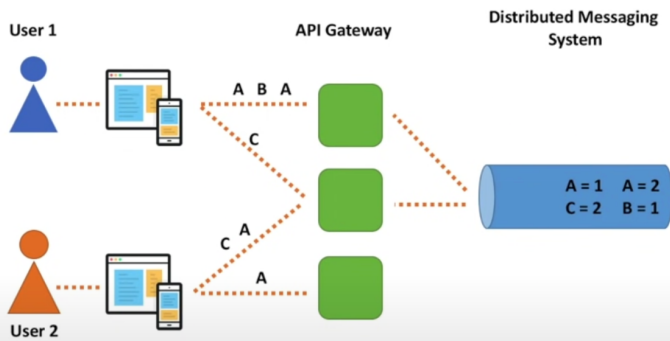
High-level architecture



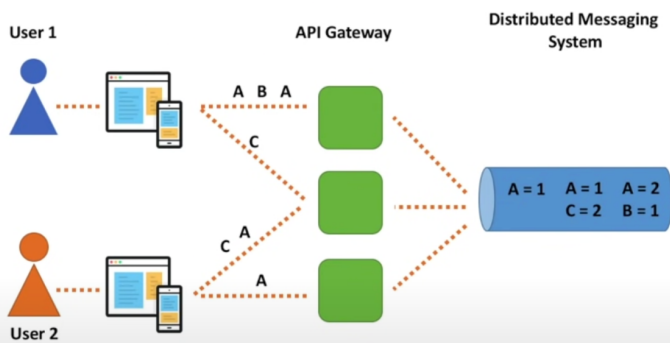
Data flow, fast path



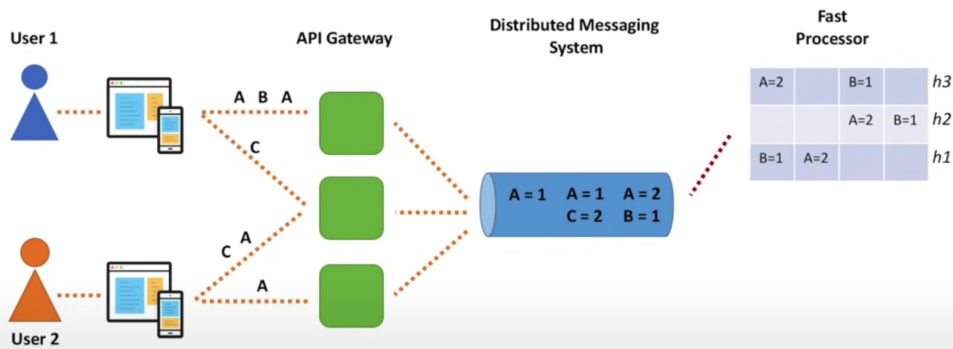
Data flow, fast path



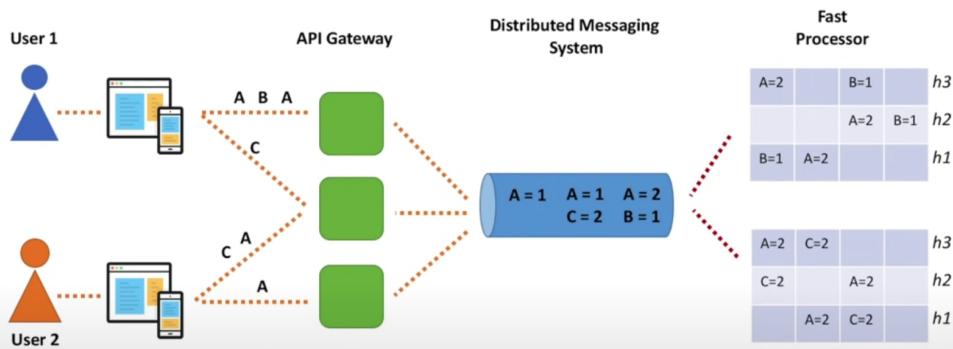
Data flow, fast path



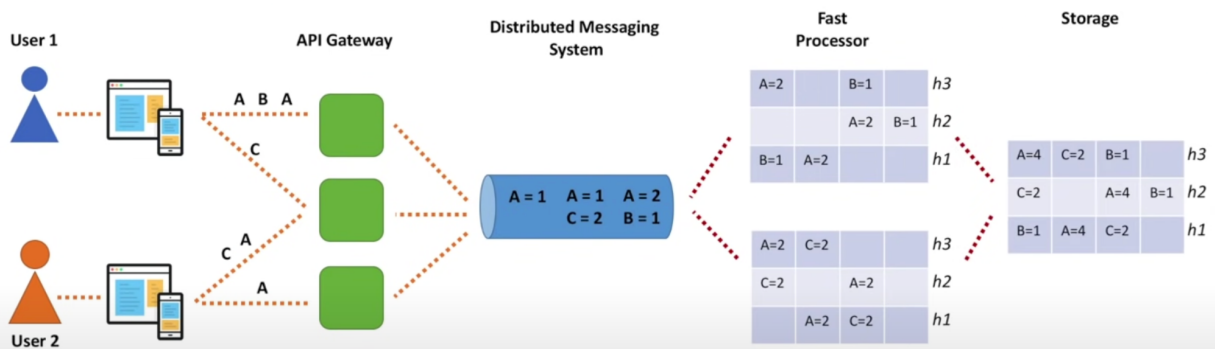
Data flow, fast path



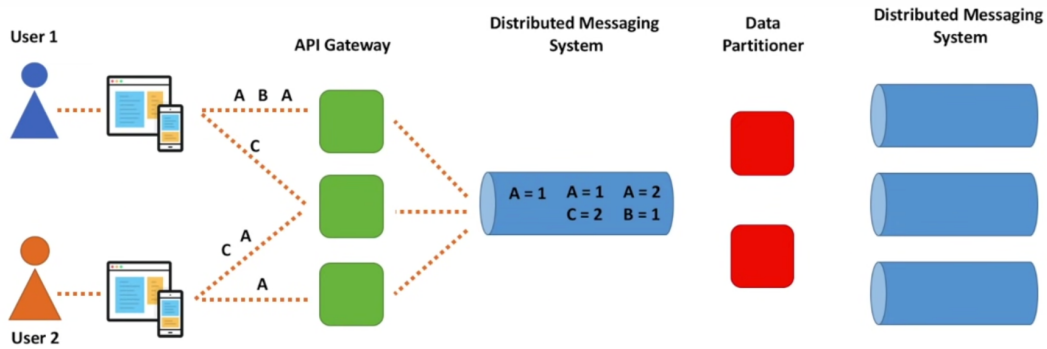
Data flow, fast path



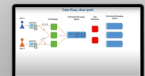
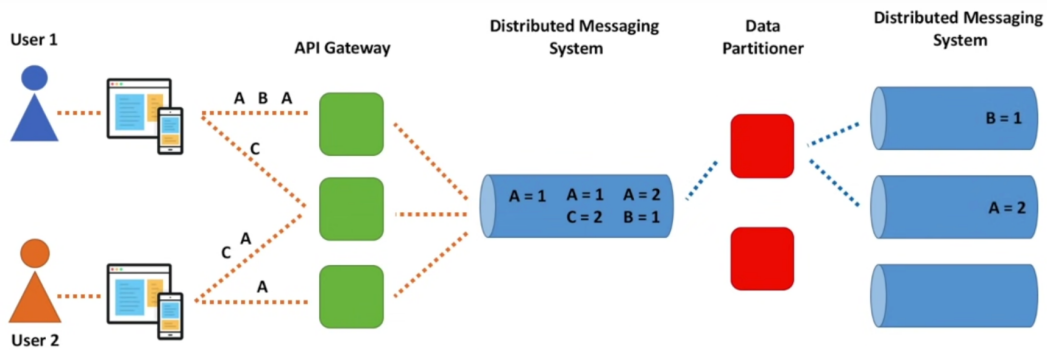
Data flow, fast path



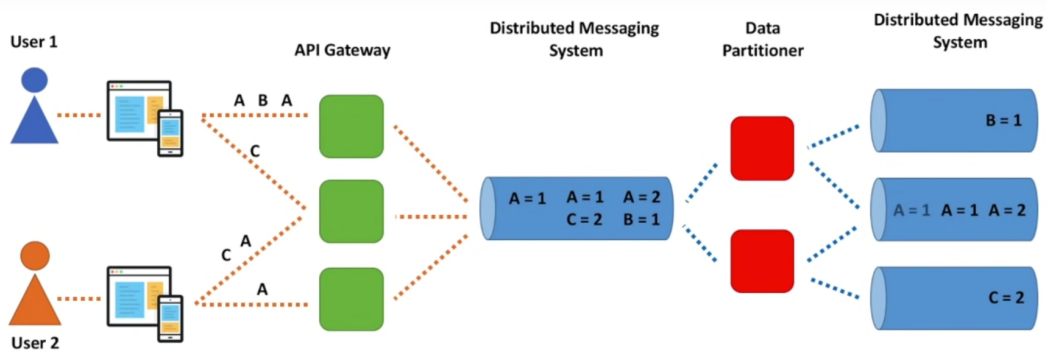
Data flow, slow path



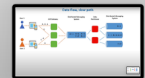
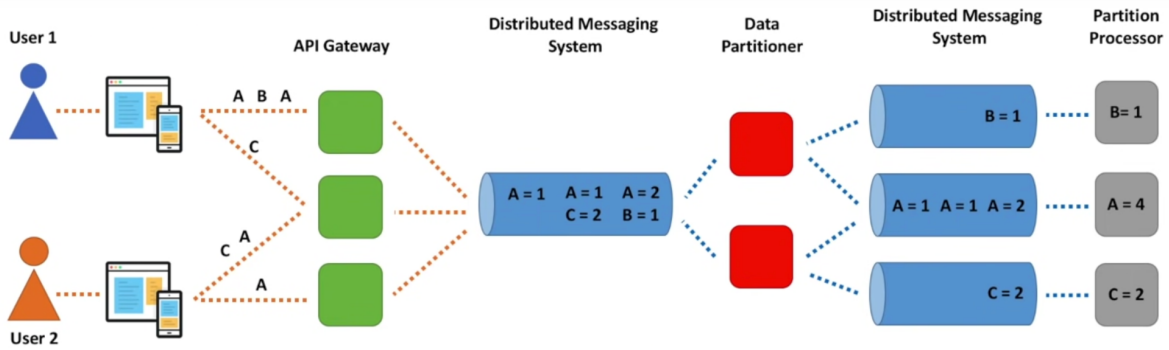
Data flow, slow path



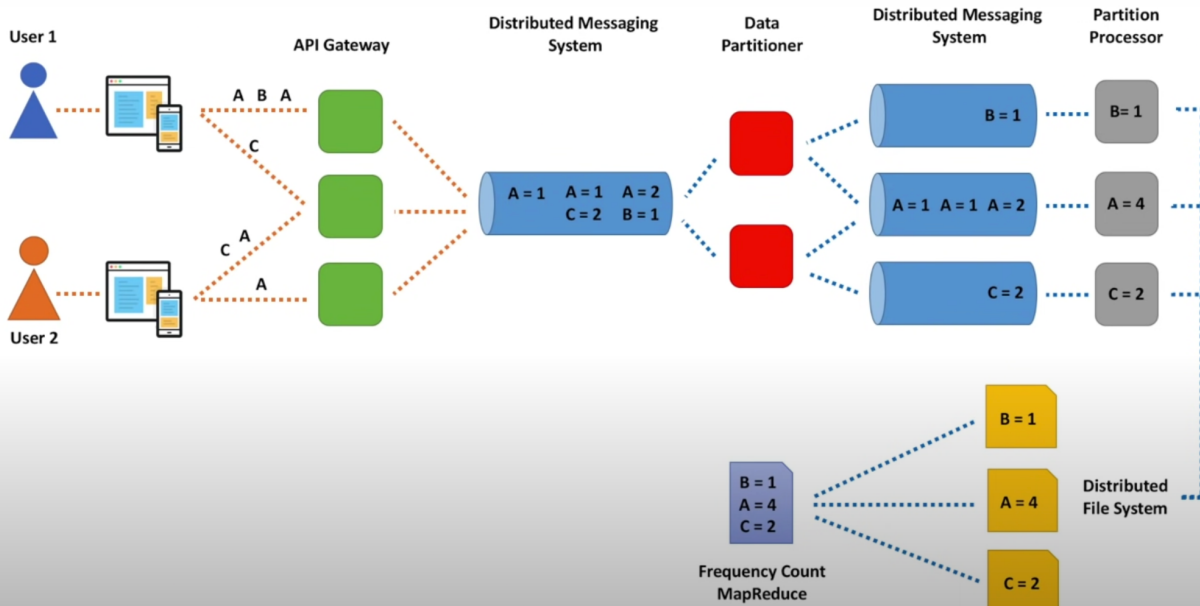
Data flow, slow path



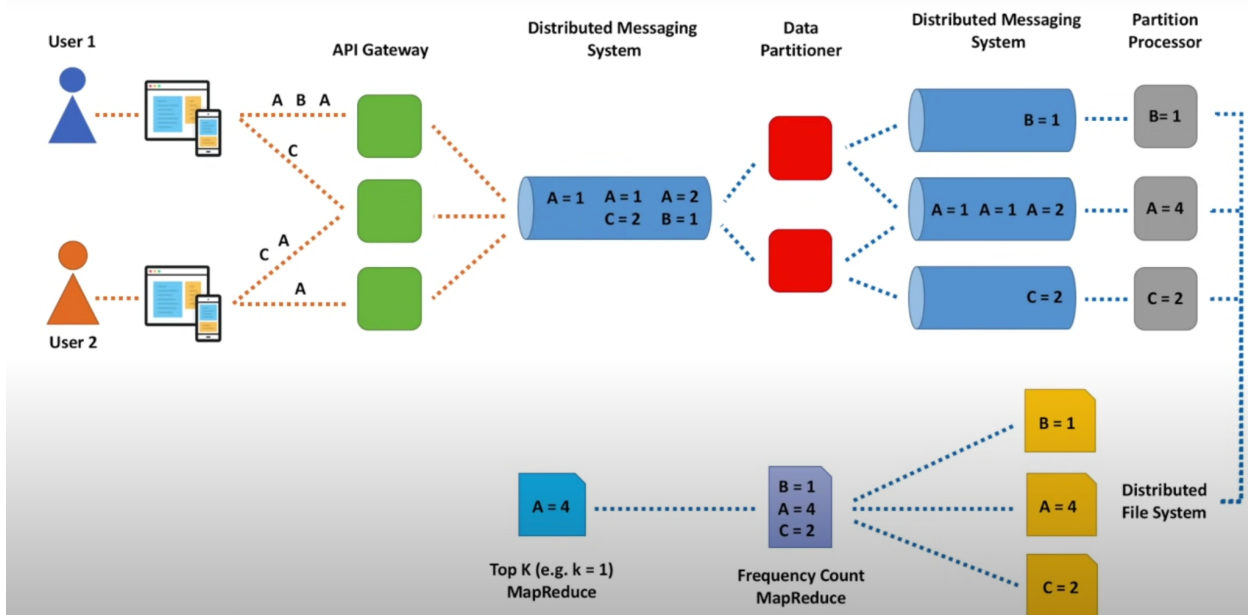
Data flow, slow path



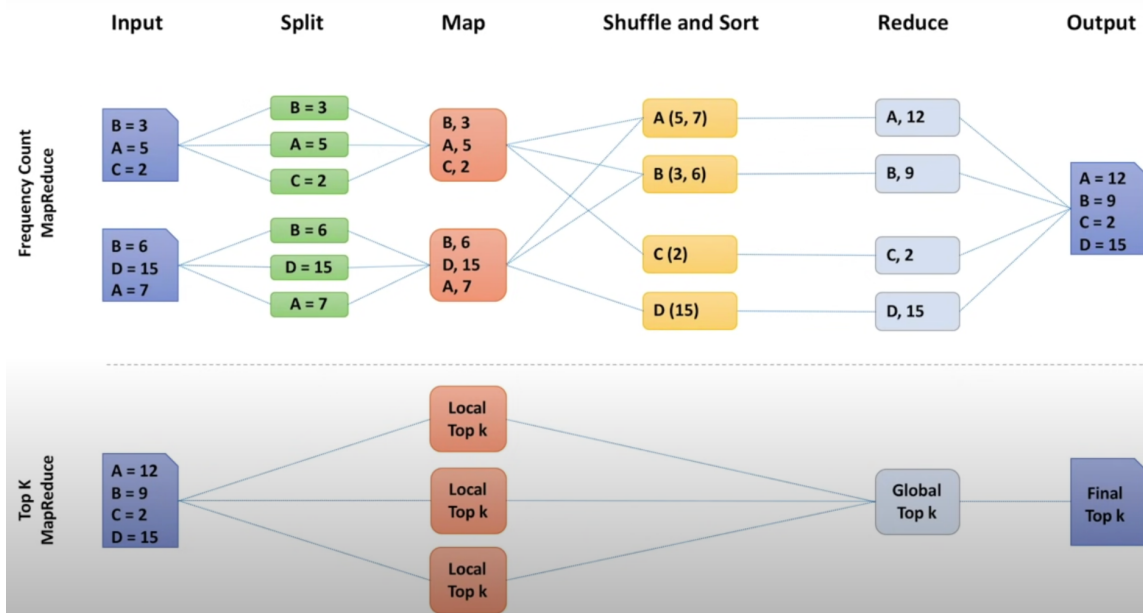
Data flow, slow path



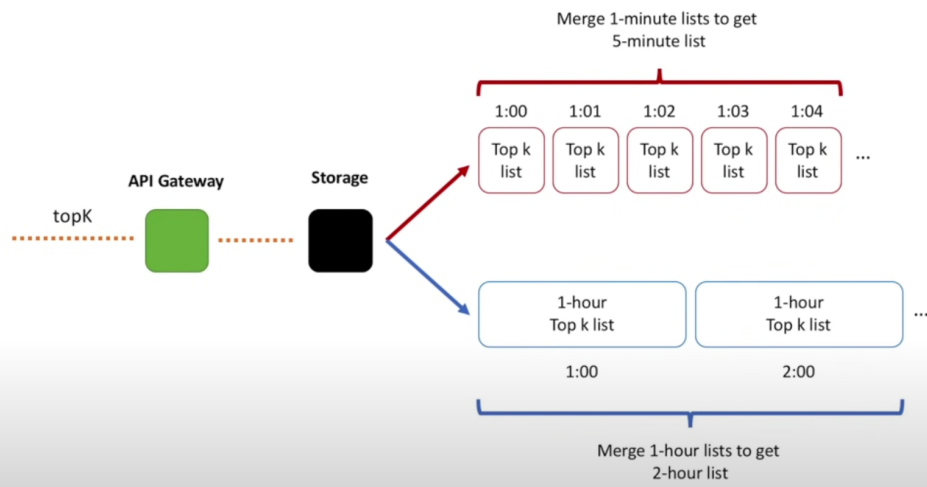
Data flow, slow path



MapReduce jobs



Data retrieval



What else can the interviewer ask?

You mentioned we aggregate data on the API Gateway side. What if this is not possible due to CPU or memory constraints on API Gateway hosts?

Any alternatives to count-min sketch?

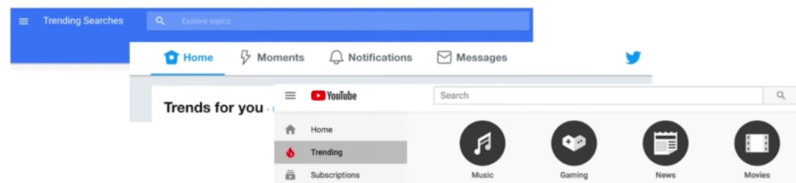
How big is the k ?



What are the drawbacks of this architecture?

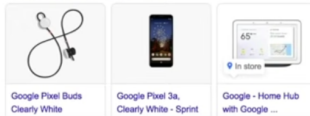
Solution to the top k problem can be also applied to...

Trends



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Market Movers

Most Actives | Gainers & Losers | Orders by Fidelity Customers

Today's most active stocks on the NYSE, Nasdaq, and AMEX.

Symbol	Volume	90-Day Avg.	T. Gain
AMD	61,202.5k	71,506.4k	-
GE	59,071.7k	76,149.8k	-
CHK	54,715.1k	45,555.7k	-

Volatile Stocks



DDoS Attack Prevention